SMI SPRING MANUFACTURERS INSTITUTE

Coil Spring Analysis Software Reference Card

Extension, Compression and Torsion Springs

HAMILTON
Engineering Services

General Instructions

ON or NEW PROBLEM NEW PROBLEM specify type of problem, then
any key displays the value of that key.
INPUT any key Input, change, or free a value
press DIAG repeatedly when prompt appears
ACK acknowledge messages and unlock keybd

Use of FN and CTL shifts

FN PITCH reciprocal of pitch
CTL PITCH pitch angle
FN Pit nominal available initial tension
FN Material define custom material type
FN DIAG recap analysis (see user manual)
FN PRINT print entire design to PC-324
FN R1 hook bending stress at R1
FN R2 hook torsional stress at R2
FN ID or OD or Body Len value at max defl
FN F1 or F2 COMP: defl @ load / defl @ solid
FN F1 or F2 or Fmax EXT: free length / defl
FN Sit S1 or S2 or Smax stress / MTS

Notes and Reminders

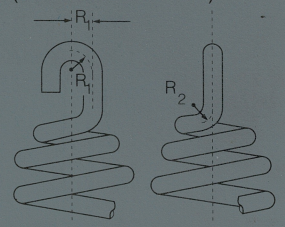
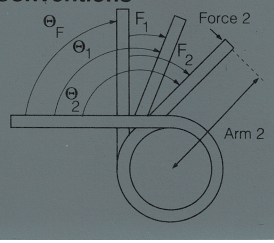
Enter or calculate wire size early.
Enter parameters, not design constraints.
Rectangular: t always less than b.
DEAD coils are in ADDITION to those already dead for compression ends.
2 standard hooks (N +2) added to extension DL only if FEED is NOT specified.
Auto Power Down does not work in numeric input or CALC mode.
INPUT, FN, and CTL shifts may be pressed either before, or concurrently with the desired shifted key. Press the shift a second time to turn off shift-mode if desired.

Torsion End Handling

FEED total amount to add to DL for legs
ENDS total ACTIVE length of legs.
ARM1, ARM2 X FORCE1, FORCE2 = M1, M2

Hook Stress Definitions

(evaluated at Pmax)



Torsion Spring Conventions